# Do Now · retrieval

## Synoptic Revision 2 — Final Test Preparation

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

### Q1

A username must contain at least one character. Give an example of invalid input and explain what the program should do.

### Q2

For a full adder with A = 1, B = 0 and CarryIn = 1, calculate Sum and CarryOut.

### Q3

A loop starts with total = 0 and adds the values 2, 4 and 6 once each. What is total after the third iteration?

### Q4

A half adder has inputs A and B. Its Sum output is A XOR B and Carry output is A AND B. For inputs (A,B) = (0,1), (1,0) and (1,1), calculate both outputs and explain why the final case produces a carry.





# Exit Ticket · reflect and retrieve

## Exit Ticket

Can you name the two exact things you will check before handing in the test?

## Use key vocabulary from memory

1. Explain normalised in your own words.

2. Explain pointer in your own words.

3. Explain LIFO in your own words.

## Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.

# New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

## Core idea 1

Do not copy corrections — CODE:
1. Attempt from memory
2. Check mark scheme
3. Write why the mark was lost
4. Retry without looking — The retry proves the correction has stuck.

## Annotate the key visual

Explain which labels or features show the main process.





# You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 2

Personal checklist
Write the two exact things you will check before handing in the test next lesson.

Check that you have used the technical terms and explained your decisions.

# You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 1

Data Types — OCR practice
Complete the Teams questions on negative binary, floating point and bit shifts. Mark them immediately.

Check that you have used the technical terms and explained your decisions.





# New Learning · applying the ideas

## Vocabulary · use the correct technical terms

normalised — A floating-point form using the available mantissa precision efficiently.

pointer — A location/index/address linking to another item.

LIFO — Last In, First Out.

FIFO — First In, First Out.

## Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

# I Do · worked example

## Problem

Use the mark value

## Follow the worked method

1. Direction — CODE:
right shift

2. Amount — CODE:
two places

3. Final answer — CODE:
Shift right by two places.

## Why this method works

Identify the decision, rule or invariant that makes the method correct.





# We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

## Guided problem · TRY

Complete both parts
CODE:
result = ?
effect = ?

## DISCUSS

Compare how you reached the answer. Where could someone make an error?

## IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.